

Module 1: Transcendental Constant Definition

Certifies: $\alpha_0 = 299 + \pi/10$ to 5000 decimal digits
Machine Certificate v1.6 -- David Fox -- May 21, 2026

1. Claim

The transcendental α_0 used in S14 is defined as **$299 + \pi/10$** computed to **5000 decimal digits** via mpmath arbitrary-precision floating-point arithmetic ($\text{mp.dps} = 5000$).

2. Source Code

File: `certificates/alpha0.py`

```
from mpmath import mp
mp.dps = 5000
alpha0 = 299 + mp.pi/10
print(alpha0)
```

3. Build Environment

```
$ python3 --version
Python 3.11.14

$ pip show mpmath | grep Version
Version: 1.3.0

$ sha256sum alpha0.py
8175b0e904084156c249ccc185420aa98db982976320247a54f082442b6d1d49  alpha0.py
```

4. Raw Execution Log

First 120 digits of stdout (full output: 5000 digits):

```
299.314159265358979323846264338327950288419716939937510582097494459230781640628620
8998628034825342117067982148086513282306647093844609550582231725359408128481117450
2841027019385211055596446229489549303819644288109756659334461284756482337867831652
7120190914564856692346034861045432664821339360726024914127372458700660631558817488
1520920962829254091715364367892590360011330530548820466521384146951941511609433057
2703657595919530921861173819326117931051185480744623799627495673518857527248912279
38183011...
```

5. Cryptographic Binding

All three SHA-256 digests are computed with **sha256sum** (SHA-2, 256-bit) over the exact bytes of each artifact. The `invariants.json` file binds all three into a single manifest.

Item	SHA-256 Digest
Source <code>alpha0.py</code>	8175b0e904084156c249ccc185420aa98db982976320247a54f082442b6d1d49
Stdout <code>alpha0</code> value	63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291
<code>invariants.json</code>	abe8cf160531964ff90371794c25fa8ecbc3330ceb434ca3d69cc6597f1cc235

6. Verification Status

Check	Result
alpha0 = 299 + pi/10 computed to 5000 dps	PASS
Source SHA-256 bound	PASS
Stdout SHA-256 bound	PASS
invariants.json SHA-256 bound	PASS
Overall certificate status	VERIFIED

Reproduce: `python3 certificates/alpha0.py | sha256sum --` should match SHA_OUT above.

Repository: <https://github.com/DavidFox998/alpha0-ponti> -- Certificate generated by Machine Certificate v1.6